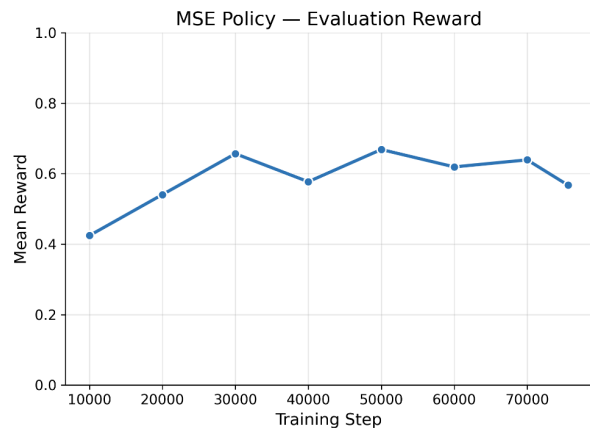
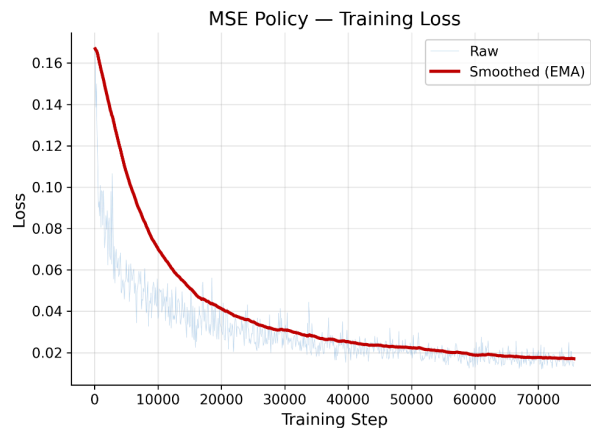


Action Chunking with MSE Loss

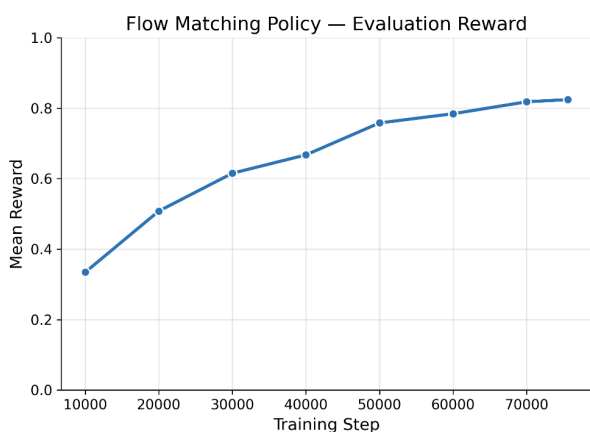
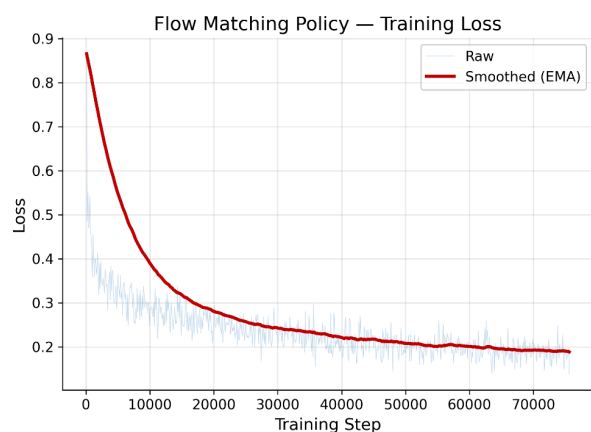


The MSE policy is a 3-layer MLP that maps the current state directly to a flattened action chunk.

- Input: State vector (R^5)
- Hidden layers: 3 FC layers with hidden size 256 each, each followed by ReLU activation
- Output: Linear layer producing $\text{chunk_size} * \text{action_dim} = 8 * 2 = 16$ values, reshaped to (batch, 8, 2)
- Loss: MSE between predicted and ground-truth action chunks
- Optimizer: Adam ($\text{lr}=3\text{e-}4$, no weight decay)
- Batch size: 128
- Training duration: 400 epochs (75,600 steps)

The training loss decreases rapidly in the first ~10,000 steps and continues to gradually decline, reaching approximately 0.015 by the end of training. The evaluation reward plateaus around 0.57–0.67, with the best evaluation reward of 0.669 at step 50,000. The reward curve isn't improving further across evaluations, which is characteristic of MSE-based policies that tend to predict the mean of multimodal action distributions.

Flow Matching



The flow matching policy achieves substantially higher reward (0.82 vs 0.67) compared to the MSE policy. This is expected because the Push-T task has a multimodal action distribution: there are multiple valid ways to push the T-block into the target. The MSE policy, which directly regresses to the mean action, tends to average over these modes, producing less decisive and sometimes contradictory motions. In the videos, the MSE policy's pushes appear more hesitant and less precise, sometimes failing to make clean contact with the T-block.

The flow matching policy, by learning a generative model that can sample from the full action distribution, produces more committed and coherent pushing trajectories. In the videos, it tends to pick a clear strategy and execute it smoothly, resulting in better alignment of the T-block with the target. The iterative denoising process allows the model to refine its action predictions, yielding sharper and more purposeful movements compared to the averaged-out behavior of the MSE baseline.